

OOPS Concept

Polymorphism :-

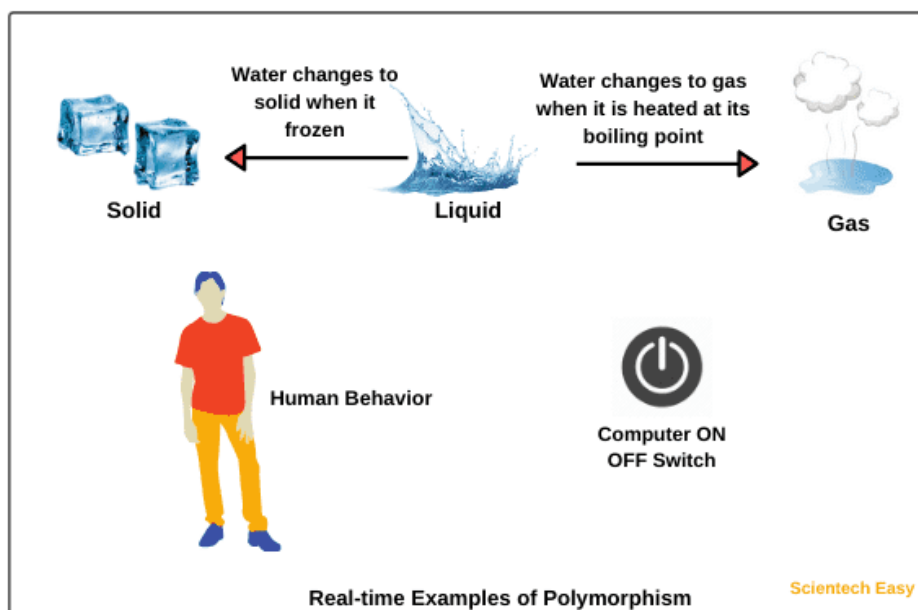
- Polymorphism in Java means one name with many forms.
- It allows the same method or object to behave differently based on the context.
- There are two types of polymorphism: compile-time (method overloading) and runtime (method overriding).
- It provides flexibility, reusability, and simplifies code in object-oriented programming.

Features of polymorphism:-

- **Multiple Behaviors:** The same method can behave differently depending on the object that calls this method.
- **Method Overriding:** A child class can redefine a method of its parent class.
- **Method Overloading:** We can define multiple methods with the same name but different parameters.
- **Runtime Decision:** At runtime, Java determines which method to call depending on the object's actual class.

Why Use Polymorphism :-

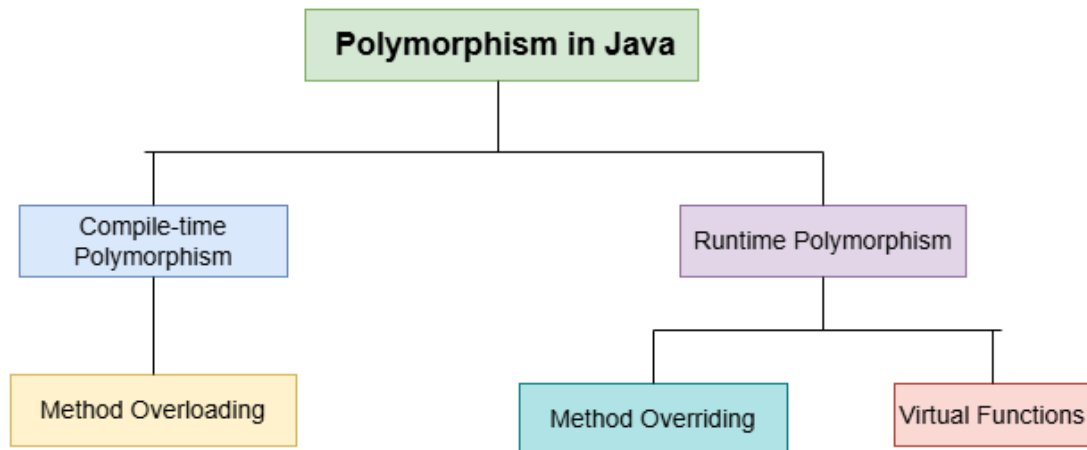
- **Code Reusability:** Same method/class works with different objects.
- **Flexibility:** Treat objects of different classes as a common superclass.
- **Abstraction:** Work with general types (superclass/interface) instead of specific subclasses.
- **Dynamic Behavior:** Java chooses the method at runtime based on the actual object.



Types of Polymorphism :-

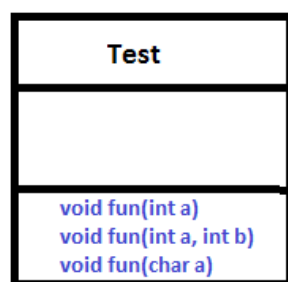
In Java Polymorphism is mainly divided into two types:

1. Compile-Time Polymorphism (Static)
2. Runtime Polymorphism (Dynamic)

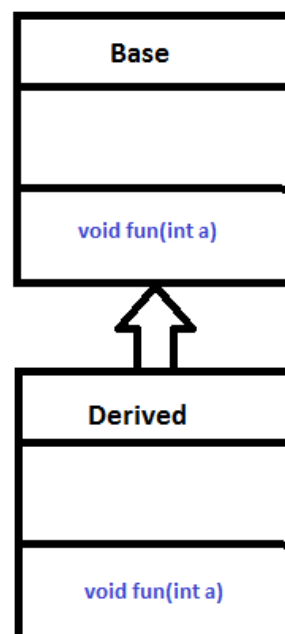


1. Method Overloading :-

- Achieved by defining multiple methods with the same name but different parameters (number or type).
- Resolved at compile time.
- Allows a method to perform different tasks based on input.



Overloading



Overriding

Method Overloading can be achieved by using:-

1) No Parameter -

```
void display() { System.out.println("No parameters"); }
```

```
void display(int a) { System.out.println(a); }
```

2) By Passing Parameter -

```
void display(int a) { System.out.println(a); }
```

```
void display(String s) { System.out.println(s); }
```

3) Changing Datatype –

```
void display(int a) { System.out.println(a); }
```

```
void display(double a) { System.out.println(a); }
```

4) Changing the Sequence of Datatype –

```
void display(int a, double b) { System.out.println(a + ", " + b); }
```

```
void display(double a, int b) { System.out.println(a + ", " + b); }
```

5) Changing the Number of Parameter –

```
void display(int a) { System.out.println(a); }
```

```
void display(int a, int b) { System.out.println(a + ", " + b); }
```

Example –

```
package oops;

public class method_overloading {
    void payment()    //by passing no parameter
    {
        System.out.println("Credit");
    }

    void payment(int num) //by passing parameter
    {
        System.out.println("Debit");
    }

    void payment(int r, String name) //by passing number of parameter
    {
        System.out.println("UPI");
    }

    void payment(String p, int c)    //by changing the sequence of parameter
    {
        System.out.println("Cash on Delivery");
    }

    void payment(float z, char w)    //by changing the datatype
    {
        System.out.println("Paytm");
    }

    public static void main(String[] args) {

        method_overloading obj=new method_overloading();
        obj.payment();
        obj.payment(1);
        obj.payment(5, "Sagar");
        obj.payment("Kate", 3);
        obj.payment(5, 'A');
    }
}
```

Output –

Credit

Debit

UPI

Cash on Delivery

Paytm

Example –

```
package oops;

public class area_task_method_overlaoding {
    void area(int l,int b)
    {
        int rectangle=l*b;
        System.out.println("Area of Rectangle is: "+rectangle);
    }

    void area(int side)
    {
        int square=side*side;
        System.out.println("Area of Square is: "+square);
    }

    void area(double r)
    {
        double circle=3.14259*r*r;
        System.out.println("Area of Circle is: "+circle);
    }

    void area(double h,double b)
    {
        double triangle=0.5*h*b;
        System.out.println("Area of Triangle is: "+triangle);
    }

    public static void main(String[] args) {
        area_task_method_overlaoding obj=new area_task_method_overlaoding();
        obj.area(5, 4);
        obj.area(3);
        obj.area(6.0);
        obj.area(7.0, 3.0);
    }
}
```

Output –

Area of Rectangle is: 20

Area of Square is: 9

Area of Circle is: 113.13324

Area of Triangle is: 10.5